

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

- Line 1 contains N1, followed by N1 integers of the first array
- Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6	10 57
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 void findIntersection() {
4     int t;
5     if (scanf("%d", &t) != 1) return;
6     while (t--) {
7         int n1;
8         scanf("%d", &n1);
9         int *arr1 = (int *)malloc(n1 * sizeof(int));
10        for (int i = 0; i < n1; i++) {
11            scanf("%d", &arr1[i]);
12        }
13        int n2;
14        scanf("%d", &n2);
15        int *arr2 = (int *)malloc(n2 * sizeof(int));
16        for (int i = 0; i < n2; i++) {
17            scanf("%d", &arr2[i]);
18        }
19        int i = 0, j = 0;
20        int first = 1;
21        while (i < n1 && j < n2) {
22            if (arr1[i] == arr2[j]) {
23                if (!first) printf(" ");
24                printf("%d", arr1[i]);
25                first = 0;
26                i++;
27                j++;
28            } else if (arr1[i] < arr2[j]) {
29                i++;
30            } else {
31                j++;
32            }
33        }
34        printf("\n");
35        free(arr1);
36        free(arr2);
37    }
38 }
39
40 int main() {
41     findIntersection();
42     return 0;
43 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.